

Hydronic Water Pressure - Field Guide

Demo knowledge document | Ref: DEMO-HYD-001 | Rev 1.0

Important: This document is fictional material created for an AI-agent demonstration. Values and procedures must be validated against the actual appliance manual, installation standard and site conditions before field use.

1. Typical sealed-system pressure

For this demo system, the normal cold fill pressure is assumed to be approximately **1.0-1.5 bar**. When the system heats, pressure may rise. A typical operating range in this fictional example is around **1.5-2.2 bar**. Pressure behaviour depends on static head, vessel charge, system volume and temperature.

2. Quick diagnosis

Observation	Likely causes to investigate	Field checks
Pressure below 0.8 bar when cold	Water loss; recent bleeding; small leak; relief-valve discharge; filling loop closed after incomplete fill	Inspect visible joints/radiators; check discharge pipe; verify gauge; top up only if permitted by appliance instructions
Pressure repeatedly drops after topping up	Leak; PRV passing; expansion-vessel issue; automatic air vent loss	Do not repeatedly mask the fault by filling. Check for discharge and loss points.
Pressure approaches 3 bar when hot	Expansion vessel lost charge/undersized; filling valve passing; vessel isolated	Compare cold/hot pressure; check vessel connection; inspect relief discharge
Pressure high even when cold	Overfilled system; filling loop passing; gauge error	Isolate filling arrangement; reduce only using approved procedure; verify gauge

3. Expansion-vessel logic

Symptom pattern

A large swing from normal cold pressure to near relief-valve opening pressure when hot is a classic reason to investigate the expansion vessel. Vessel pre-charge should only be checked with the water side depressurised and according to the appliance/system instructions.

Safety boundary

Never rely on a generic demo value for maximum pressure, relief-valve rating, vessel pre-charge or filling procedure. The AI agent should identify the appliance/model and retrieve the approved technical document before instructing a field engineer.